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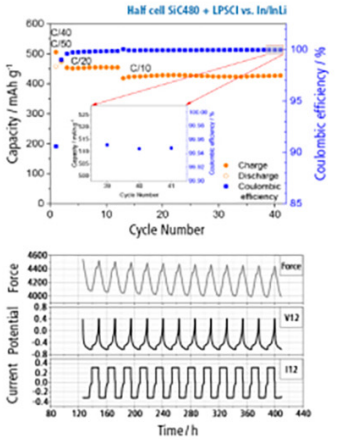
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Long-Term Study on the Impact of Depth of Discharge, C-Rate, Voltage, and Temperature on the Lifetime of Single-Crystal NMC811/Artificial Graphite Pouch Cells

A. Eldesoky,[1] M. Bauer,[2] T. Bond,[2,3] Nicholas Kowalski,[4] J. Corsten,[2] D. Rathore,[2] R. Dressler,[2] and J. R. Dahn[1,2,z]

[1]*Department of Chemistry, Dalhousie University, Halifax, NS, Canada*
[2]*Department of Physics & Atmospheric Science, Dalhousie University, Halifax, NS, Canada*
[3]*Canadian Light Source, Saskatoon, SK, Canada*
[4]*Department of Mechanical Engineering, Dalhousie University, Halifax, NS, Canada*

This work examined the impact of depth of discharge (DOD), C-rate, upper cut-off voltage (UCV), and temperature on the lifetime of single-crystal NMC811/Artificial Graphite (AG) cells. Cells were cycled at C/50, C/10, C/5, or C/3, and 25, 50, 75, or 100% DOD at room temperature (RT, 20 ± 2 °C) or 40.0 ± 0.1 °C. The UCVs were 4.06 or 4.20 V. After 12000 hr of cycling, experiments such as electrochemical impedance spectroscopy (EIS), Li-ion differential thermal analysis (DTA), ultrasonic mapping, X-ray fluorescence (XRF), differential capacity analysis, synchrotron computed tomography (CT) scans, and cross-section scanning electron microscopy (SEM) were carried out. We showed that capacity loss increased slightly with DOD and C-rate, and that cells with 4.06 V UCV have superior capacity retention and impedance control compared to 4.20 V. SEM, CT scans, and differential capacity analysis show that microcracking and positive electrode mass loss did not occur regardless of DOD, C-rate, or UCV. DTA and ultrasonic mapping showed no C-rate or DOD dependency for electrolyte changes or "unwetting." Finally, a simple square-root time model was used to model SEI growth in 4.06 V UCV cells.
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In recent years, $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NCM811) positive electrode materials and other Ni-rich formulations have been utilized by the battery industry, owing to their high energy density and lower cost.[1] Despite their popularity, high Ni cells have shorter lifetime compared to other NMC grades with lower Ni content, such as NMC532 for example, especially at high voltages,[2,3] Strategies such as electrolyte engineering,[3–5] and positive electrode coatings and/or dopants[6,7] have been utilized to improve the lifetime of NMC811 and Ni-rich cells, but Ni-rich cells operated to 4.20 V continue to show inferior lifetime regardless of advances in electrolytes or material engineering. Battery cell degradation depends on factors such as temperature, charge-discharge current (C-rate), depth of discharge (DOD), and upper cut-off voltage (UCV), all of which will impact the cell capacity and resistance over time. Therefore, understanding the impact of all these factors on the lifetime of Ni-rich Li-ion cells is critical to maximizing their lifetime, which will result in superior energy storage products and lessen the burden of battery recycling.

Numerous publications examined the impact of cycling conditions on the cycle life of various cell chemistries.[8–17] For example, it has been demonstrated that charging at higher currents and/or lower temperatures induced Li plating,[11,12] and a large body of work agrees that capacity fade increases with SOC and/or DOD.[14,16,17] In addition to cycle life, studies on the impact of temperature and SOC on calendar life of NMC/Graphite cells have been carried out in various commercial cell formats.[18–20] For example, Gao et al. studied the impact of SOC on the capacity retention of unspecified 8 Ah NMC/Graphite prismatic cells and demonstrated that loss of positive electrode active material was a prominent contributor to cell failure at high SOCs and is very small when the cell is cycled between 0%–20% SOC.[14] Similar observations were made by Gauthier et al. in polycrystalline NMC622/Natural graphite cells where positive electrode mass loss increased at higher DODs after ~2.5 years of cycling.[9] In contrast, previous work from our group has demonstrated the positive electrode active mass loss and microcracking is virtually eliminated in NMC materials when single-crystal morphologies are used.[21] Additionally, the use of artificial vs natural graphite materials greatly impacts the rate of parasitic reactions and Li loss,[22,23] and aging studies on commercial cell formats do not always disclose which type of graphite material is being used. Lastly, aging studies on commercial cells rarely involve probing the electrode and electrolyte changes of aged cells, so using convenient pouch cell formats is valuable since it allows us to easily investigate chemical and physical changes to the cell. Despite the differences in cell design among various aging studies, some key similarities arise:

1. Cycling at higher rates and/or low temperatures can result in Li plating;[12,18]
2. Generally, higher DODs and SOCs result in increased capacity fade and impedance growth;[14,16,17]
3. Li inventory loss is the prominent source of capacity loss and positive electrode active mass loss can contribute to cell failure when particle cracking is prominent;[9]
4. Higher temperatures exacerbate Li inventory loss and cell degradation.[18,20,24]

In this work, we studied the impact of DOD, C-rate, UCV, and temperature on capacity retention and impedance growth of single-crystal NMC811/Kaijin AML400 artificial graphite pouch cells with an EC:EMC:DMC electrolyte blend with a 2% VC and 1% DTD additive mixture. Cell and electrolyte specifications can be found in Table SI. This work uses cell and electrolyte chemistries which demonstrated excellent cycle life,[2] and a single-crystal NMC811 material which has been shown to not exhibit any micro-cracking.[21] Figure 1 summarizes the design of the experiment performed here. The UCV was fixed to either 4.06 or 4.20 V and the lower cut-off was varied according to the DODs shown in Fig. 1. The 4.06 V UCV was chosen to avoid the large lattice collapse that occurs at high voltages (see Fig. 1). The C-rates used here were C/50, C/10, C/5, or C/3, and the DOD choices were 25, 50, 75 or 100% DOD. Cells were cycled at ambient room temperature (RT) or at 40.0 ± 0.1 °C for 12000 hr (approximately 1.5 years) followed by a range of characterization experiments: electrochemical impedance spectroscopy (EIS), differential thermal analysis (DTA), ultrasonic mapping, X-ray fluorescence (XRF), differential capacity

[z]E-mail: Jeff.dahn@dal.ca

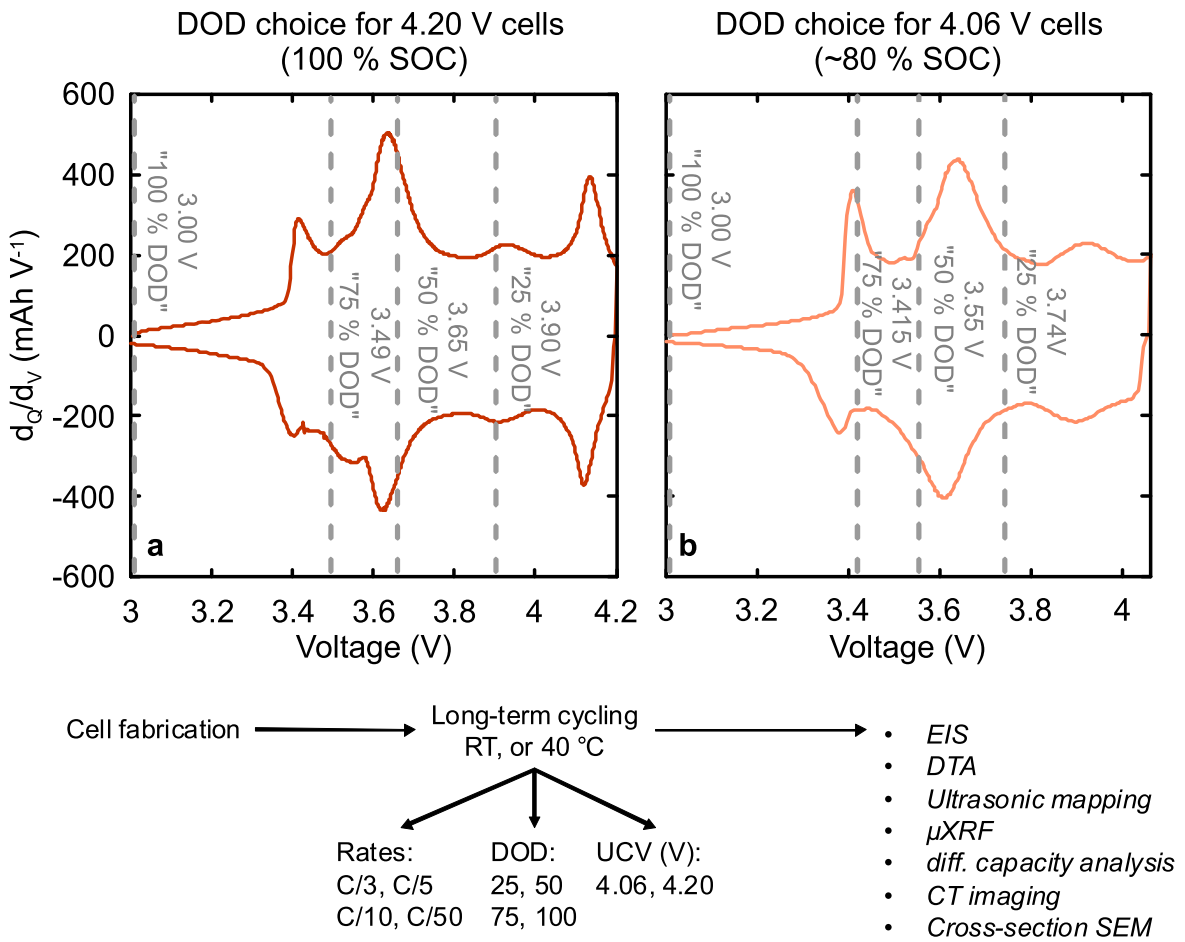


Figure 1. dQ/dV (mAh V^{-1}) vs voltage (V) for NMC811/AG cells cycled to (a) 4.20 or (b) 4.06 V UCV, and a summary of the experiments carried out in this work. Dashed lines indicate the lower voltage cut-off for 25, 50, 75 and 100 % DODs used here.

analysis, synchrotron computed tomography (CT) imaging, and cross-section scanning electron microscopy (SEM). This extensive aging study on cells with demonstrated excellent cycle life, and subsequent characterization of aged electrode and electrolytes, allows for optimization of cell performance to accommodate various applications and understanding the impact of aging conditions on the physical and chemical changes in the cell.

Experimental Methods

Pouch cells and electrolyte.—Machine-made pouch cells with ~240 mAh nominal capacity were received vacuum sealed from LiFUN Technologies (Hunan province, China) with no electrolyte. The positive electrode material was single crystal NMC811, and the negative electrode was Kaijin AML400 artificial graphite. Cells were cut open in an Ar-filled glovebox and dried at 120 °C under dynamic vacuum prior to filling with 0.85 ml (~1 g) of electrolyte. The electrolyte used was a solvent blend of ethylene carbonate (EC), ethyl methyl carbonate (EMC), dimethyl carbonate (DMC) in a 25:5:70 wt. ratio and 1.5 M $LiPF_6$ salt. The additive blend was 2wt% vinylene carbonate (VC) and 1 wt% ethylene sulfate (DTD). Cells were held at 1.5 V for 24 h after electrolyte filling to allow for proper wetting of the jelly roll, then underwent a C/20 charge-discharge formation cycle between 3.0 to 4.06 or 4.20 V at 40.1 ± 0.1 °C. After the formation cycle, cells were charged to 3.80 V for electrochemical impedance spectroscopy (EIS) as described below. Cell and electrolyte specifications are listed in Table SI.

Electrochemical impedance spectroscopy (EIS).—EIS spectra were measured at 10.0 ± 0.1 °C and 3.8 V (~50% SOC) using a Biologic VMP3 multi-channel potentiostat. Data was collected from 100 kHz to 10 mHz with a signal height of 10.0 mV. Ten data points were collected per decade in frequency. Positive electrode area specific impedance is reported throughout.

Long-term cycling.—Long-term cycling was done on Neware cyclers (Shenzhen, China). Cells were cycled at ambient room temperature (20 ± 2 °C) or 40.0 ± 0.1 °C between 3.0 and 4.06 or 4.20 V. C-rates used were C/50, C/10, C/5, or C/3, and DODs of 25, 50, 75, or 100% as shown in Fig. 1. Capacity check cycles were done periodically at 100% DOD and C/10:C/10 to probe the cell capacity. Table SII contains a list of all cells used in this work along with their cycling conditions. Researchers interested in the raw data can reach out to the authors with the "Cell IDs" of interest.

Differential capacity analysis.—Before and after cycling, two C/40:C/40 cycles were recorded at 40 °C (3.0–4.20 V) on Dalhousie's ultra-high precision chargers for use in differential capacity analysis. The cells first did a C/40 discharge to 3.0 V, followed by 12 hr hold at 3.0 V to "drain" the Li in the 1 mm anode overhang region, then full C/40 cycles at 3.0-4.20 V. Differential capacity analysis was done as previously described.[25,26]

Synchrotron computed tomography (CT) scans.—Computed tomography (CT) scans were carried out at the Canadian Light Source as described by Bond et al.[27]

Differential thermal analysis (DTA).—Li-ion Differential Thermal Analysis (DTA) is an *in-situ* method for determining the state of an electrolyte in a cell. This method functions by heating a

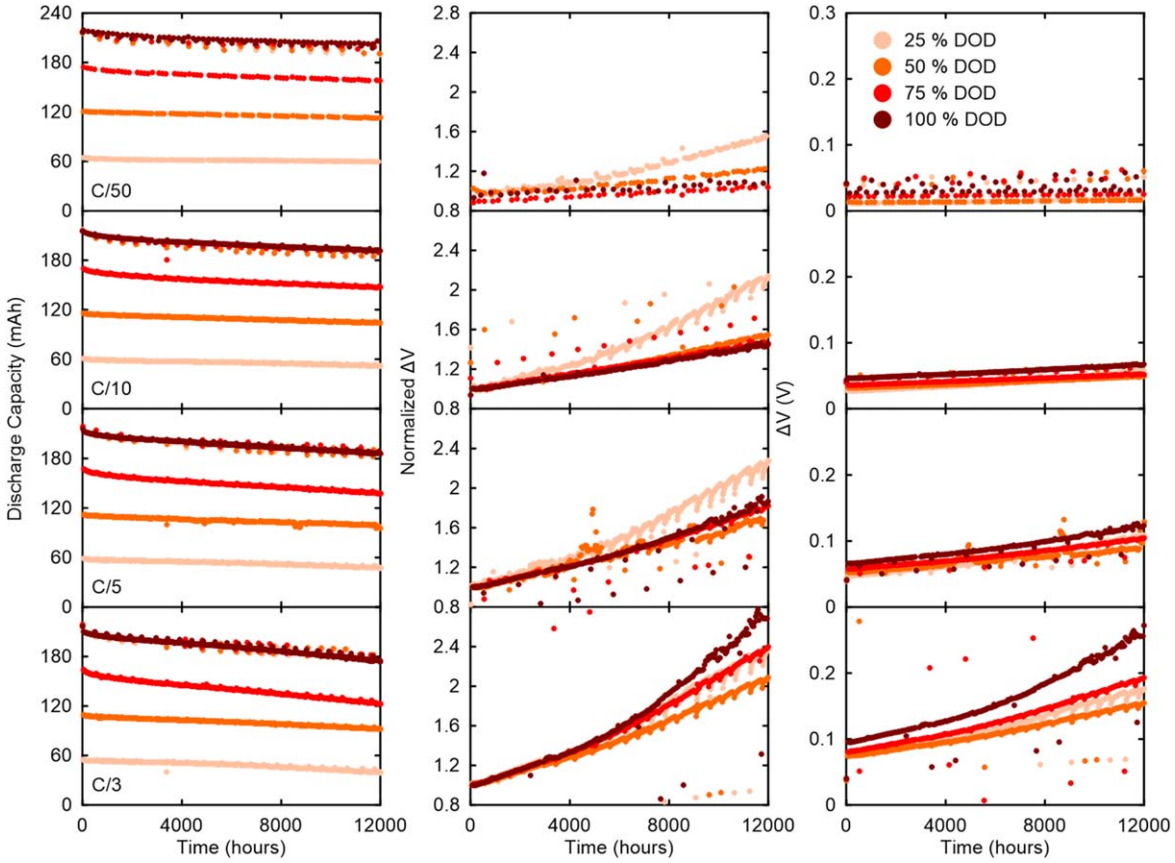


Figure 2. Discharge capacity (mAh), normalized ΔV, and absolute ΔV (V) vs time (hours) for NMC811/AG cells cycling to 4.20 V UCV at 40 °C at 25, 50, 75, and 100% DOD. The C-rate is indicated in each panel.

cell at a controlled rate from cryogenic temperatures, where the electrolyte in the cell is fully frozen, to ambient temperatures, where it is fully liquid. Deviations from the controlled heating rate indicate phase changes, and these phase changes can subsequently be used to determine the state of the electrolyte in the cell in situ.[28,29]

Ultrasonic transmission mapping.—Ultrasonic transmission mapping was performed using an ultrasonic battery scanner (UBSC-LD50, Jiangsu Jitri-Hust Intelligent Equipment Technology Co., Ltd) as described by Deng et al.[30] A pair of ultrasonic focus transducers were positioned on either side of the cell. The transducers and cells were immersed in low-viscosity silicone oil that serves as an ultrasonic coupling agent. The transducers were installed on a two-dimensional motion system with a precision of 0.2 mm to perform mechanical progressive scanning. The ultrasonic signal was emitted by a transducer on one side of the cell and received by the transducer on the other side. The waveform of the transmitted signal was recorded with a collecting card. The peak-to-peak values of the received transmission waves were converted into colour heat maps to produce the pseudo-colour images.

X-ray fluorescence (XRF).—XRF scans on negative electrodes extracted from cycled cells were done to quantify transition metal dissolution and deposition on the negative electrode as described by Eldesoky et al.[31]

Cross-section scanning electron microscopy (SEM).—Cross-section SEM images were taken on positive electrodes from cycled pouch cells. Cells were charged to 4.20 V and opened in an Ar-filled glove box. Positive electrodes were rinsed with DMC twice before ion-milling with a JEOL IB19530CP cross-section polisher. Electrodes were milled with an Ar-ion beam for a 75 min and 6 kV coarse step followed by a 5 min and 4 kV fine step. SEM images were taken using a Hitachi S-4700 field emission electron microscope with a secondary electron detector. The images were obtained using an accelerating voltage of 7 kV and current of 20 μA.

Results

Long-term cycling data.—Single-crystal NMC811/AG pouch cells were cycled at 40 °C or RT with UCVs of 4.20 or 4.06 V and discharged to various DODs as described in Fig. 1. Cells were cycled at C-rates of C/3, C/5, C/10, and C/50 with periodic 100% DOD, C/10:C/10 capacity check cycles. Pair cells were cycled and one of each pair was stopped after ~8600 hr (1 year) of cycling for various characterizations, while the other cell continued cycling for another ~4000 hr. Figure 2 shows the discharge capacity (mAh), normalized ΔV, and absolute ΔV (V) vs time for cells with 4.20 V UCV cycled at 40 °C, and the capacity check cycles are shown separately in Fig. S1. The RT counterpart to Fig. 2 is shown in Fig. S2. The C-rate is indicated in each panel and the color intensity increases with the DOD choice. ΔV here is taken as the difference between the average charge and discharge voltages, which increases with the cell's internal resistance. The right column of Fig. 2 shows $\Delta V/\Delta V_{10}$, that is ΔV value normalized by ΔV measured at the 10th cycle. Fig. 2 shows that the ΔV growth increases as the C-rate increases which suggests that more volume changes per unit time leads to more rapid impedance growth.

Figure 3 shows the discharge capacity (mAh), normalized ΔV, and absolute ΔV (V) vs time for cells with 4.06 V UCV cycled at 40 °C, and the RT counterpart is shown in Fig. S3. One apparent and dramatic difference between Figs. 2 and 3 is the minimal ΔV growth for the 4.06 V UCV cells compared to their 4.20 V counterparts. For example, ΔV for 4.20 V cells cycled at C/3 at 40 °C more than doubled after 12000 hr of cycling (Fig. 2) while the 4.06 V C/3 cells showed a ~25% increase at the most. This dramatic reduction in

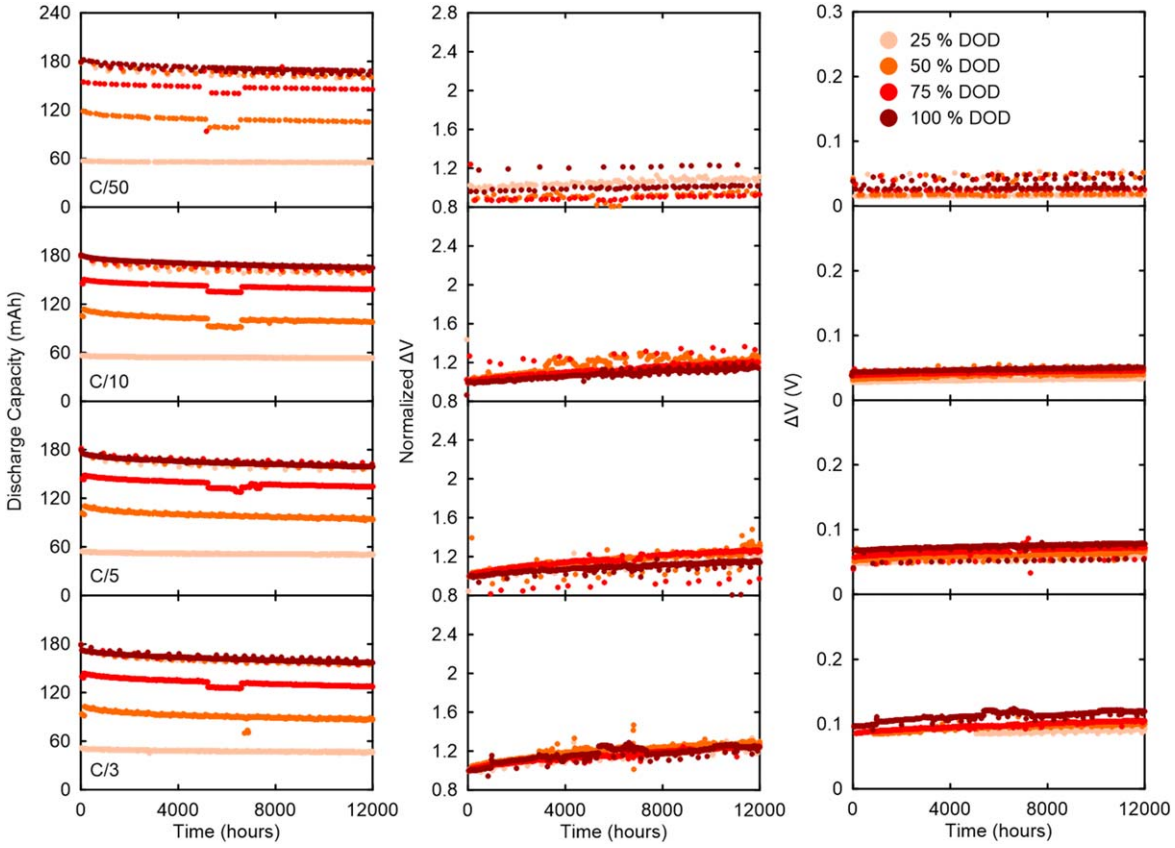


Figure 3. Discharge capacity (mAh), normalized ΔV, and absolute ΔV (V) vs time (hours) for NMC811/AG cells cycling to 4.06 V UCV at 40 °C at 25, 50, 75, and 100 % DOD. The C-rate is indicated in each panel. The dip in capacity around ∼5500 hr is due to failure of one 40 °C temperature chamber after a power interruption where the cells tested at ambient temperature for some time.

impedance growth for NMC811/graphite cells tested to 4.06 V has been reported before by Song et al.[3] and by Eldesoky et al.[23] Additionally, the 4.06 V cells in Fig. 3 show a significantly improved capacity retention compared to the 4.20 V cells.

Figure 4 shows the absolute capacity loss (mAh) at t = 12000 hr from the C/10:C/10, 100% DOD, capacity check cycles vs C-rate for all cells in this work. Figure S4 shows the same data plotted vs DOD. Cells cycled at 40 °C show greater capacity loss than their RT counterparts as expected. Additionally, there is an overall increase in capacity fade with increasing current which can be attributed to the larger cycle number of C/3 cells vs C/50 cells, for example. Apart from Fig. 4a, it is worth noting that larger DODs generally resulted in *smaller* capacity loss as is clear in Figs. 4b–4d. Capacity loss for cells cycled at 40 °C with a 4.20 V UCV at C/3 and C/5, however, showed no DOD dependency. Overall, differences between the various DODs in Fig. 4 were within 10–15 mAh which is small. Looking at Fig. S4, it is clear that cycling at larger DODs and faster rates results in more capacity fade. For example, Fig. S4a shows that a 4.20 V cell cycled at C/3 and 100% DOD lost ∼12 mAh more capacity than a C/50 100% DOD one. Interestingly, the impact of C-rate on capacity loss in Fig. S4a becomes more significant at higher DODs.

Figure 5 shows the percent growth in ΔV for all cells in this work at t = 12000 hr vs C-rate. Figure S5 shows the same data plotted vs. DOD. Figures 5a–5b highlight the dramatic impact of UCV on ΔV growth, where the 4.20 V cells demonstrated up to a 200% increase in ΔV and the 4.06 V counterparts showed no more than 25% increase. Additionally, the impact of temperature on ΔV growth is shown in Figs. 5c–5d where minimal ΔV growth (<30%) was seen after 12000 hr of cycling at RT regardless of UCV. We believe the differences shown in Figs. 5a–5b are quite remarkable and highlight a persistent issue with Ni-rich cells. Similarly, Fig. S5 shows that ΔV growth is minimal for all cells except ones cycled at 4.20 V UCV and 40 °C (Fig. S5a). Figure S5a also shows that ΔV growth increases with C-rate for all DODs.

***Post-cycling analysis.*—**After 1 year of cycling, one of each pair cell was removed from cycling to undergo further analysis with EIS, XRF, ultrasonic mapping, and DTA, while the second cell continued cycling for another ∼4000 hr. Figure 6 shows EIS results after cell formation and after 1 year of cycling at 4.06 or 4.20 V UCV at 40 °C for the cells with different DODs and C-rates in this work. EIS spectra were collected at 50% SOC and 10 °C, and the positive electrode-area specific impedance is reported here. All cycled cells showed an increase in charge transfer resistance (R_{CT}), simply defined as the diameter of the "semi-circle" in the Nyquist plot in this case, compared to the post-formation results as expected. However, the changes in R_{CT} in the 4.06 V cells showed no C-rate- or DOD-dependency; in fact, all 4.06 V cells showed a nearly identical R_{CT} after 1 year of cycling at 40 °C. On the other hand, the 4.20 V cells showed that R_{CT} increases with both DOD and C-rate. However, at 25% DOD the current dependence of R_{CT} for the 4.20 V cells after cycling is negligible.

A commonly proposed failure mechanism in NMC cells is transition metal (TM) dissolution and deposition on the negative electrode. Here, we used X-ray fluorescence as described previously[31] to quantify the amount of Ni, Mn, and Co deposited on the negative electrode extracted from cells cycled for 1 year at 40 °C. Figure 7 shows the concentration of Ni, Mn, and Co (μg cm^{-2}) detected on negative electrodes taken from 4.20 and 4.06 V cells cycled at 40 °C for 1 year, compared to levels of TM detected after formation. Overall, insignificant amounts of TMs were found on the negative electrodes of these cells, which is expected for a robust single-crystal NMC material where TM dissolution and

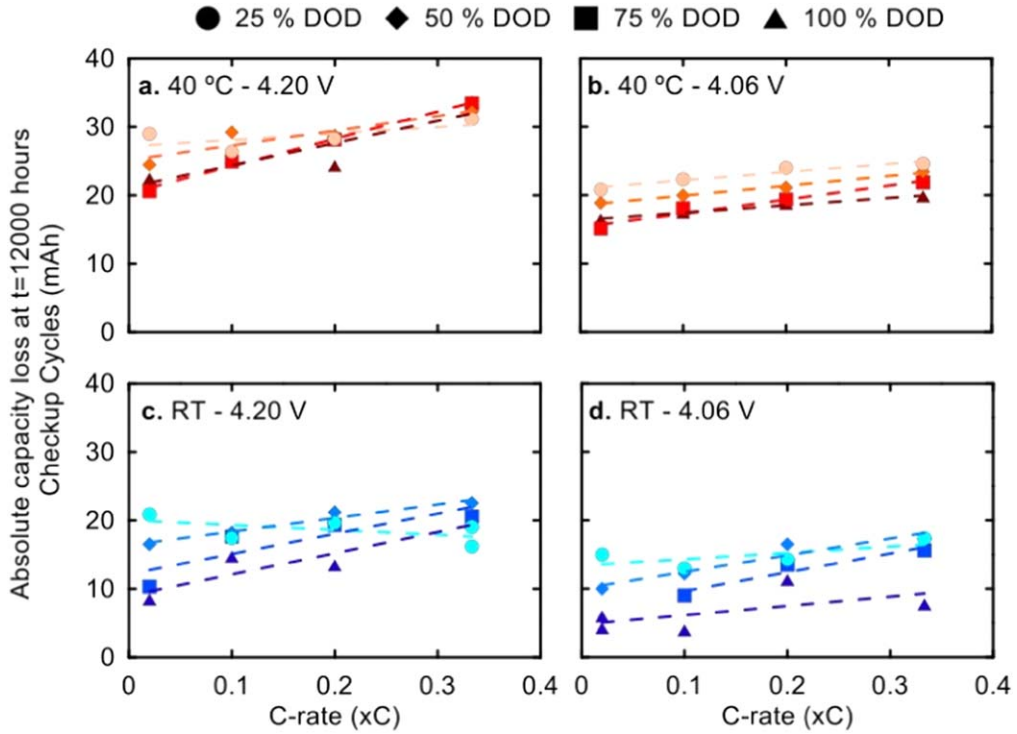


Figure 4: Capacity loss (mAh) from the 100 % DOD, C/10:C/10 capacity check cycles at t = 12000 hr vs C-rate for all cells in this work. (a)–(b) show the capacity loss for the 40 °C cells cycled to 4.20 and 4.06 V UCV, and (c)–(d) show the capacity loss for the equivalent RT cells. The different DODs are indicated by the different symbols.

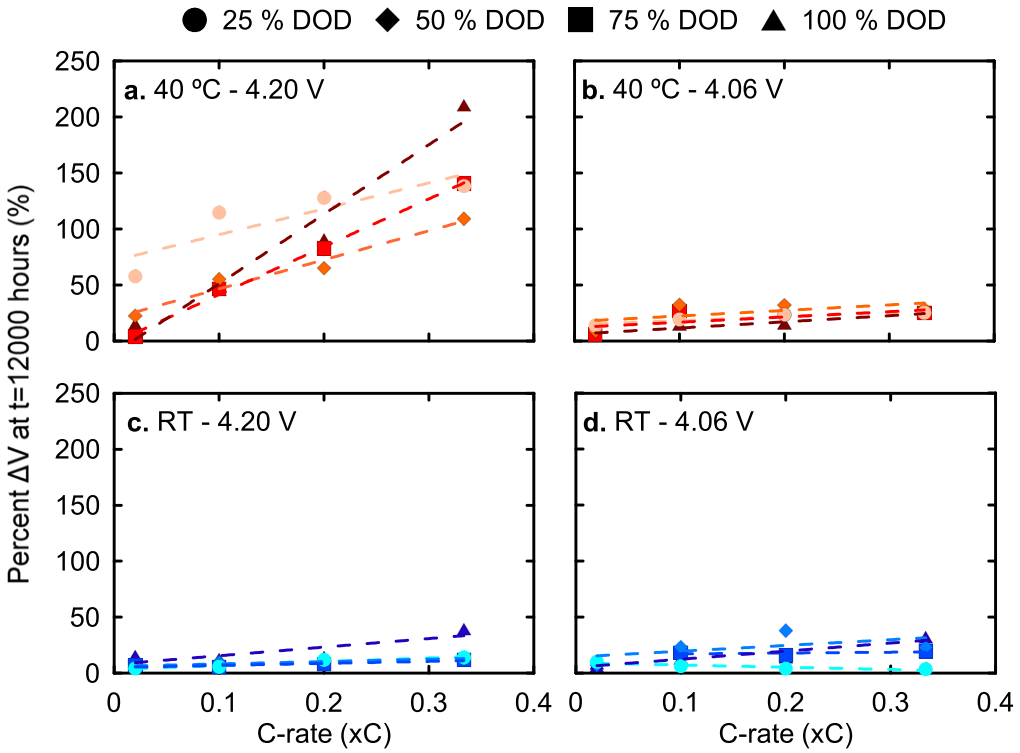


Figure 5. Percent ΔV growth at t = 12000 hr vs C-rate for all cells in this work. (a)–(b) show the ΔV increase for the 40 °C cells cycled to 4.20 and 4.06 V UCV, and (c)–(d) show the ΔV increase for equivalent RT cells. The different DODs are indicated by the different symbols.

deposition is not an issue.[32–34] For example, Thompson et al. showed that [Mn] found on the negative electrodes of NMC532 cells cycled at UCVs of 4.0 to 4.40 V at 55 °C was between 1–3.5 μg cm^{-2} (<0.1% of total Mn in the positive electrode), which did not significantly contribute to capacity loss.[32]

Previous work from our group studied the impact of DOD and C-rate on the lifetime of poly-crystalline NMC622/Natural graphite cells, where it was shown that large cell volume change resulted in electrolyte uptake due to the increased electrode porosity, which resulted in poor ultrasonic transmission.[9] Here, we used ultrasonic

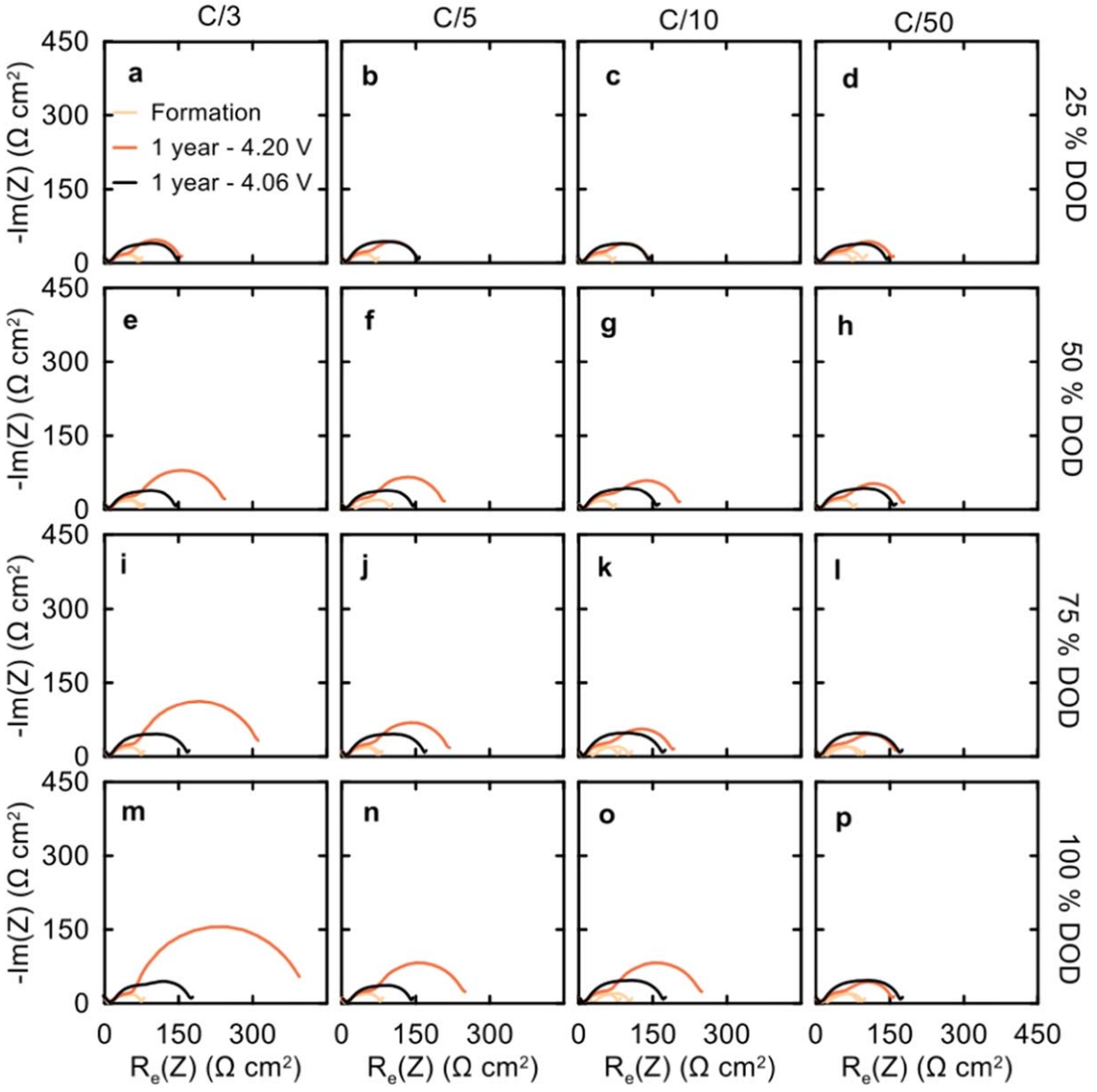


Figure 6. Nyquist plots collected after formation and 1 year of cycling for NMC811/AG cells cycling to 4.06 or 4.20 V UCV at 40 °C at (a)–(d) 25, (e)–(h) 50, (i)–(l) 75, and (m)–(p) 100 % DOD. C-rates are indicated for each column. All EIS spectra were recorded at 10 °C and 3.80 V. The EIS spectra were recorded between 100 kHz (at the left side) to 10 mHz (at the right side).

mapping to examine electrolyte distribution in 4.20 V cells cycled at 40 °C for 1 year to probe if electrolyte uptake and jelly roll 'unwetting' had occurred. Figure 8 shows the ultrasonic maps obtained on 4.20 V UCV cells after 1 year of cycling at 40 °C, where the rows and columns indicate the various C-rates and DODs, respectively. High-to-low transmission is shown using a red-to-blue heat map. The blue, low transmission, regions indicate insufficient electrode wetting, while the green indicates sufficient electrolyte wetting on the jelly roll. Figure 8 shows that electrolyte unwetting was not an issue after 1 year of cycling at 40 °C and 4.20 V UCV. For examples of cells with severe electrolyte unwetting resulting in low ultrasonic transmission, we point the readers to some of our previous work using this technique.[30,35]

In addition to electrolyte unwetting, the electrolyte of aged cells can undergo chemical transformations that are triggered by solvent and salt consumption.[36,37] Various strategies to probe chemical changes to the electrolyte have been used. Here, DTA was used because it is a non-destructive measurement with high sensitivity to changes in solvent and/or salt ratios in the cell.[28] Figure 9 shows the DTA curves of cells cycled at 40 °C for 1 year with a 4.20 V UCV. The peak occurring at −20 °C corresponds to the liquidus point of the electrolyte-composition phase diagram which is sensitive to salt concentration.[28,33,38] The curves in Fig. 9 show that there is no significant C-rate- or DOD-dependency to any of the electrolyte changes in these cells, suggesting that the electrolyte in the cells shown in Fig. 9 underwent similar chemical transformations regardless of the cycling conditions at 4.20 V UCV, which is consistent with observations made by Gauthier et al.[9] Compared to a cell that only underwent formation (Fig. 9, black dashed line), the changes in the DTA curve after cycling are minimal in general, meaning that the electrolyte did not dramatically change after 12000 hr of cycling at 40 °C. While the shift in the liquidus temperature after cycling, relative to the formation data, suggests a slight increase in salt concentration presumably due to solvent consumption, changes to solvent composition could also result in changes to the liquidus point temperature.[28,33] Therefore, it is difficult to accurately state if the small changes to the liquidus point are due to solvent consumption or chemical transformations

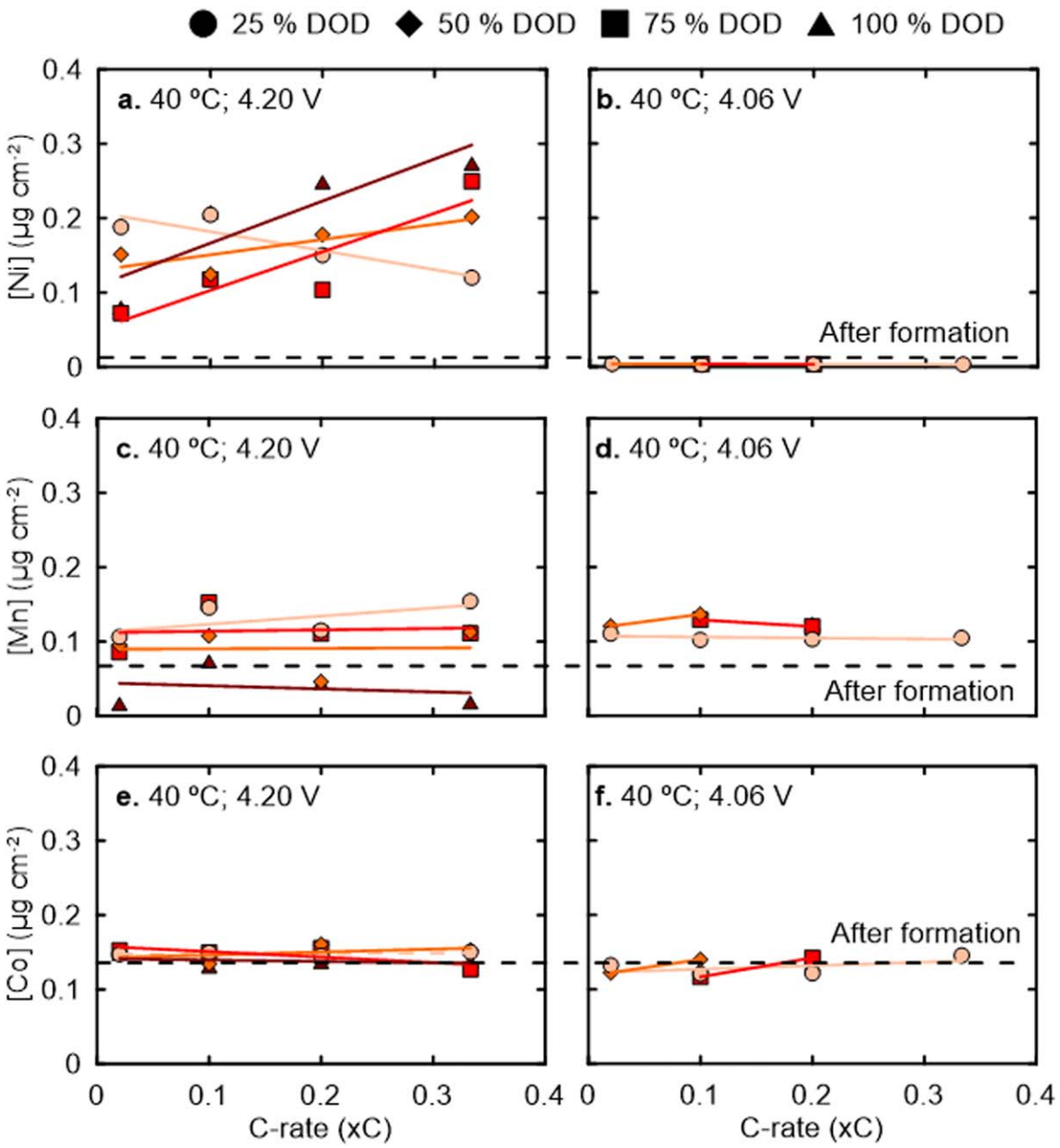


Figure 7. (a)–(b) [Ni], (c)–(d) [Mn], and (e)–(f) [Co] (μg cm^{-2}) on graphite electrodes extracted from NMC811/AG cells cycled 40 °C for 1 year vs C-rate. Dashed lines indicate [TM] after formation only, and the DODs are indicated with different symbols.

involving the solvent species, such as transesterification or EC consumption.

Differential capacity analysis and physical changes.— Differential capacity analysis is a useful tool to deconvolute the contributions of Li inventory loss (or "shift loss") and/or active material loss to capacity fade. Figure 10 shows the calculated positive and negative electrode voltage curves and the measured full cell voltage curve after formation (t = 0 hr) and cycling (t = 12000 hr). The cells in Fig. 10 were cycled at 40 °C with a 4.20 V UCV at C/3 and different DODs. The full cell charge curve has zero capacity at the point where the full cell is discharged. Figure S6 shows excellent agreement between the measured and calculated dV/dQ vs Q curves, and additional differential capacity analysis results are shown in Figs. S7–S13. Figure 10 shows that relative electrode slippage increased significantly with cycling which is consistent with Li inventory loss, and the 25 and 50% DOD cells showed more relative slippage compared to the 75 and 100% DOD counterparts. Differential capacity analysis was performed on a representative set of cells to highlight the impact of DOD, C-rate, and UCV on inventory loss and active material loss, if any, and the results are shown in the SI and summarized below.

Figure 11 shows positive electrode active mass and shift loss vs DOD or C-rate for cells cycled at 40 °C under different conditions as shown in the panels. It is clear in Figs. 11a and 11c that there is no significant change in positive electrode active mass after prolonged cycling. This observation is consistent with the robustness of single-crystal NMCs to particle cracking.[21] Figure 11b shows that shift loss decreases slightly with DOD, from ~22 mAh at 25% DOD to 18 mAh at 100% DOD; and Fig. 11d shows that shift loss increases with UCV but is relatively unchanged from C/50 to C/5 but increases at C/3.

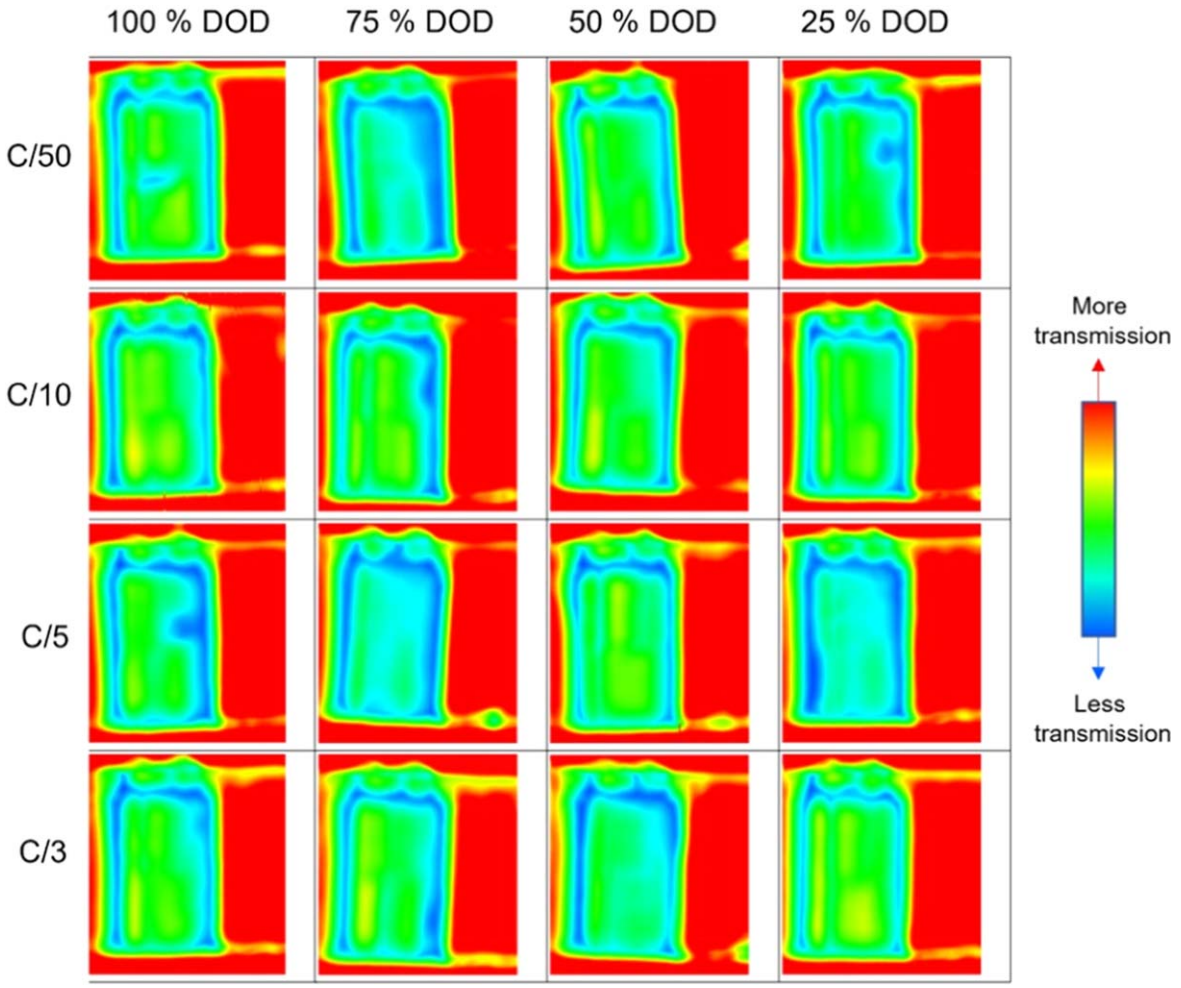


Figure 8. Ultrasonic transmission mapping of NMC811/AG pouch cells cycled for 1 year at 40 °C and 4.20 V UCV. The different columns and rows indicate the different DODs and C-rates, respectively.

To further confirm the absence of positive electrode active mass loss observed in differential capacity analysis, cross-section SEM and synchrotron CT scanning were done. Figure 12 shows images of cells dissembled at top of charge and cross-section SEM images of the positive electrode. The cells in Fig. 12 were cycled at 4.06 or 4.20 V UCV at 25% DOD with a C/5 rate at RT. The 25% DOD cells were imaged here because the choice of lower cut-off voltage in these cells results in increased cycling over the so-called remnant of the "H2-H3" phase transition region in NMC811 for the 4.20 V cell (see Fig. 1). Figure 12 remarkably shows that after 12000 hr of cycling, cells cycled to 4.06 and 4.20 V UCV at RT show virtually no separator discoloration or signs of Li plating on the anode. Further, cross-section SEMs of the positive electrode showed no signs of particle microcracking which is a known advantage in single-crystalline NMC particles. SEM images of pristine electrodes can be found in Ref. 21.

In addition to cross-section SEM, synchrotron CT was used to demonstrate the robustness of the single-crystal NMC811 against microcracking. Figure 13 shows 2D cross sections taken from NMC811 cells cycled at 40 °C with a 4.20 V UCV for 1 year. The different DODs and C-rates are indicated in the panels and a fresh comparator cell was scanned as received from our supplier. In all cases, Fig. 13 (and Fig. S15) shows that there were no significant changes to the electrode thicknesses after 1 year of cycling, which, alongside Fig. 12, further supports our differential capacity analysis results which showed no positive electrode mass loss. Table SIII lists the anode and positive electrode thickness from CT images in Fig. 13, where it is very clear that thickness changes after cycling is insignificant compared to the fresh cell. By contrast, work by Gauthier et al.[9] and Bond et al.[27] on polycrystalline NMC622/graphite cells cycled for 2.5 years at 100% DOD showed significant positive electrode active mass loss, significant microcracking and large positive electrode thickness increase.

Modelling SEI growth.—Having shown that capacity loss in the cells used in this work is primarily due to Li inventory loss after 12000 hr of cycling under different conditions, it became important to understand how the different cycling conditions impact SEI growth. In cells with minimal impedance growth, such as the cells cycled to 4.06 V in this work (Fig. 3), capacity loss can be described by the familiar square-root time model of SEI growth:[39–42]

$$Q(t) = Q_o(1 - A\sqrt{t}) \quad [2]$$

where $Q(t)$ is capacity at time t, Q_o is initial capacity, and A is a proportionality term related to the rate of SEI growth. To determine the A term as a function of C-rate and DOD, the 100% DOD C/10 capacity check cycles for all the 40 °C, 4.06 V UCV, cells were fitted with equation [2] (see Fig. S16 for some fitting examples and Table SIV for all the fitting results). Figure 14 shows the extracted A parameter vs C-rate for cells cycled at 25, 50, 75, 100% DOD at 4.06 V UCV and 40 °C. Figure 14 shows that cycling at higher rates and *smaller* DODs results in greater rate of SEI growth. Moreover, comparing the results in this work to our previous study on the

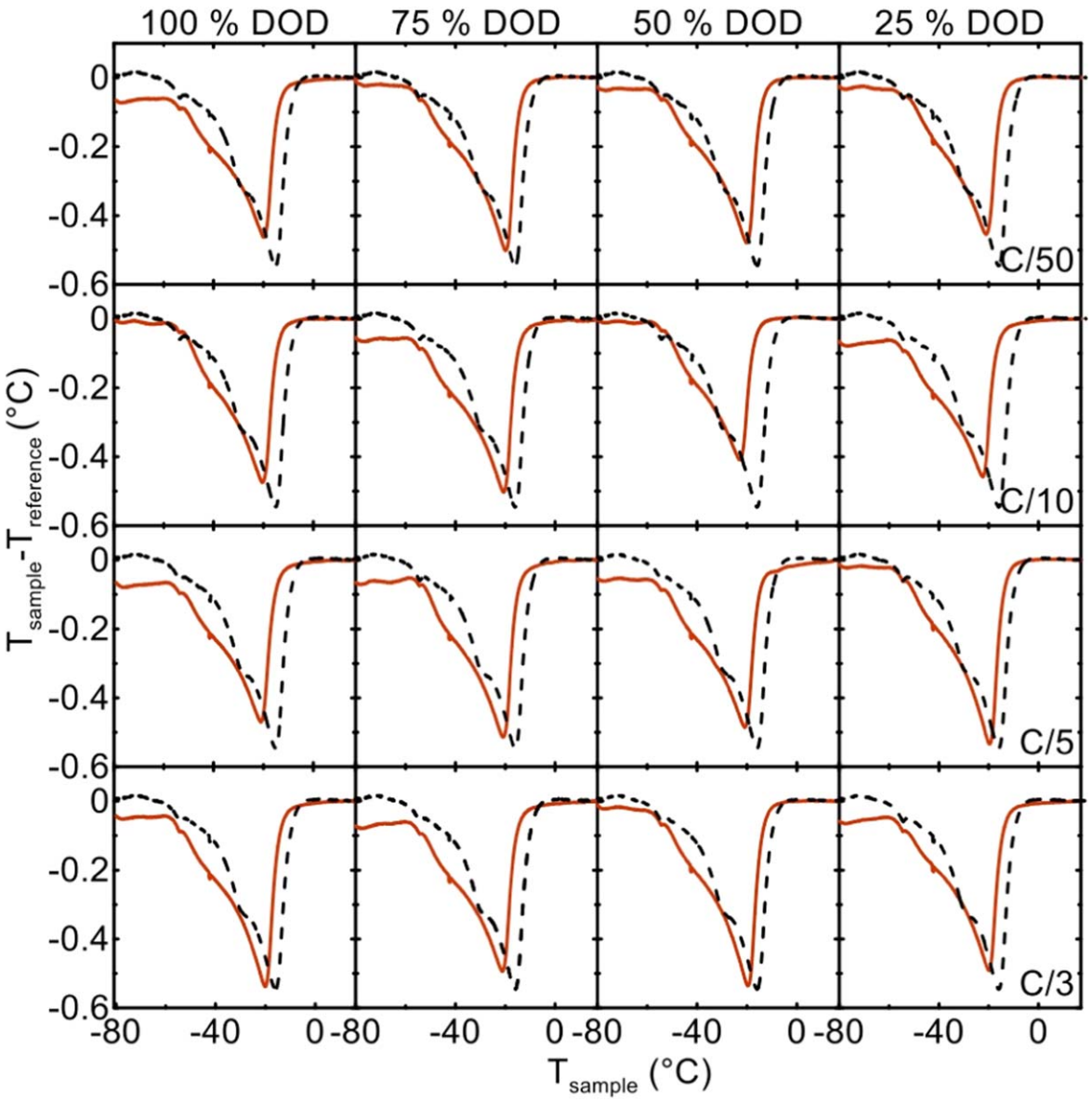


Figure 9. DTA curves of NMC811/AG pouch cells cycled for 1 year at 40 °C and 4.20 V UCV. The different DODs and C-rates are indicated in the columns and rows, respectively. The dashed line is the DTA curve of an identical cell after formation only.

impact of graphite materials on the lifetime of NMC811 cells shows that the rate of SEI growth can be reduced further using improved graphite materials that have a smaller electrochemically active surface area, which results in improved capacity retention compared to the Kaijin AML400 material used in this work.[22,23]

So far, we have examined the role of C-rate and DOD on the lifetime of NMC811/AG cells and demonstrated the dramatic impact of UCV on impedance growth and capacity retention. Additionally, we showed that NMC811 materials used in this work are robust against mechanical/physical degradation and that Li inventory loss dominates the capacity loss. Further, Fig. 14 shows that improved graphite materials can slow down the rate of SEI growth. To further demonstrate the impact of limiting the UCV and using improved graphite materials (such as "AGC" artificial graphite from our previous work[23]) on the cell lifetime, NMC811/AGC cells were cycled at 4.04, 4.06, 4.08, and 4.10 V UCV at 70 and 40 °C. Figure 15 shows the discharge capacity (mAh), normalized capacity, and normalized ΔV vs cycle number for NMC811/AGC cells with different UCV. The NMC811 electrode for this cell type is a mixture of single and polycrystalline particles as described previously.[23] Here, we see the dramatic impact of UCV on both capacity retention and impedance growth accelerated by cycling at 70 °C. Despite the small difference of 60 mV between the 4.04 and 4.10 V cells, the 4.04 V cell lost 10% capacity and its impedance increased by 5% at 70 °C after 400 cycles, compared to 15% capacity loss and 30% impedance growth at 4.10 V UCV.

Discussion

In this work, we examined the role of DOD, C-rate, UCV, and temperature on the lifetime of NMC811/AG pouch cells with a competitive electrolyte blend, as well as chemical and physical changes to the electrolytes/electrodes after cycling. In Figs. 2 and 3, cells were cycled to an UCV of 4.20 or 4.06 V, respectively, where the drawback of Ni-rich materials was made clear by the dramatic ΔV growth during cycling between 3.0 and 4.20 V compared to 4.06 V cycling. NMC811 materials experience a significant lattice volume collapse past ~4.06 V, often attributed to "H2-H3" phase transition, which in fact does not exist in NMC811 as demonstrated by Märker et al.[43] The dramatic lattice collapse occurring at high potentials could result in anisotropic lattice expansion and subsequent particle cracking coupled with reactivity of electrolyte with

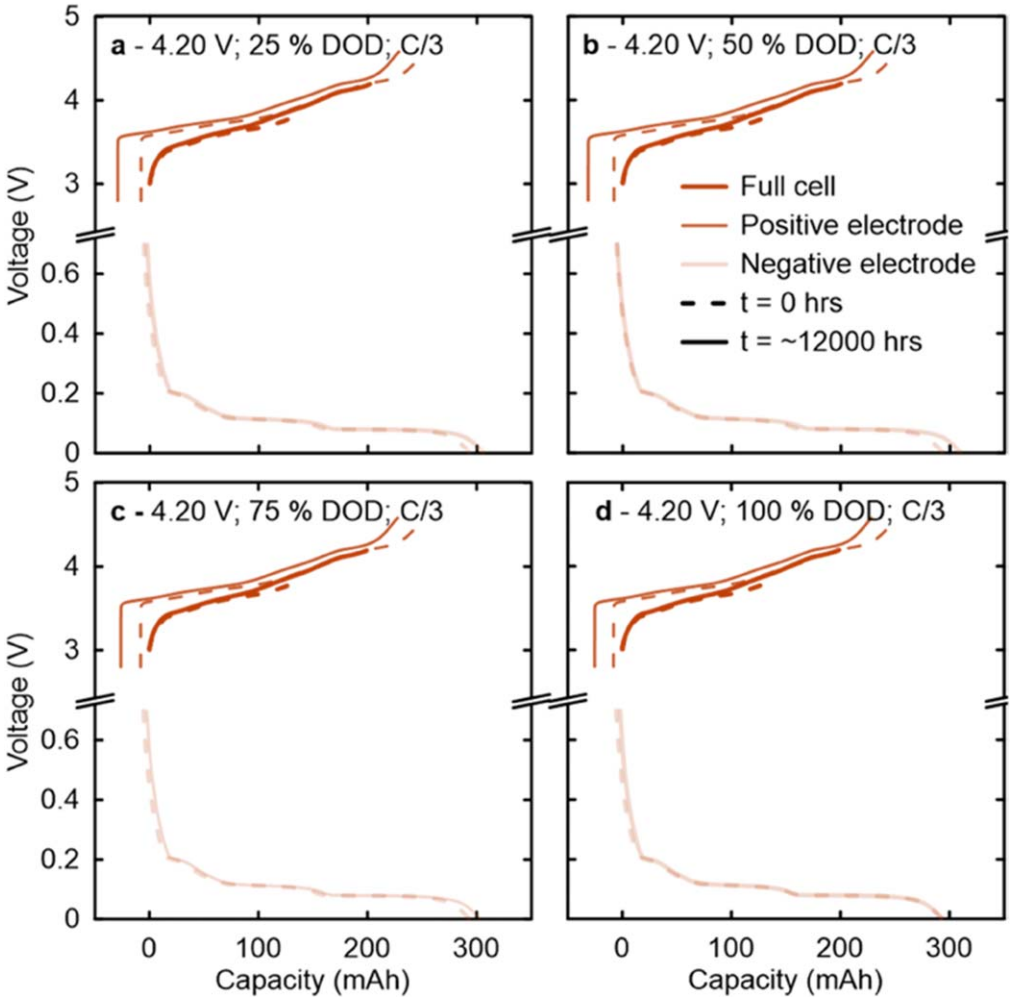


Figure 10. Voltage (V) vs capacity (mAh) curves of the full cell, negative, and positive electrodes obtained from differential capacity analysis at t = 0 (dashed line) and 12000 hr (solid line) for (a) 25, (b) 50, (c) 75, (d) 100 % DOD cells. All cells shown here were cycled at C/3 with 4.20 V UCV at 40 °C. Comparison of the measured and calculated dV/dQ vs Q curves is shown in Fig. S6.

freshly exposed positive electrode surfaces, however, this effect may be minimal in single crystal materials. Jung et al. have also shown that the NMC811 surface begins to evolve oxygen which reacts with the electrolyte during this voltage plateau above 4.06 V in a full cell.[44] Additionally, lattice collapse can decrease Li mobility and thus the accessible capacity of the cell.[43] While many approaches such as positive electrode dopants and coatings have been used to address these well-known issues[6,45,46] with varying degrees of success, we believe Fig. 3 provides a simple and economical approach: lowering the UCV to avoid lattice collapse and accelerated parasitic reactions. This approach certainly comes with an initial energy penalty, but we have previously shown that the superior lifetime of NMC811 cycled to 4.06 V vs 4.20 V can result in more lifetime energy output.[23] Eventually after prolonged cycling at 40 °C (beyond the 12000 hr shown here), the capacity of the 4.20 V cells would dip below that of the 4.06 V ones. Our previous work showed that this crossover occurs after ~5500 hr at 55 °C (C/3 cycling, 100% DOD).[23] Further, the impedance growth in the 4.20 V UCV cells (Fig. 2) will impact the power capability of these cells, while the 4.06 V ones would have relatively unchanged power capability.

Impact of DOD and C-rate on Lifetime and R_{CT}.—Figures 4 and 5 show that cycling at higher rates resulted in more capacity loss and larger impedance growth overall. Additionally, cycling at smaller DODs resulted in an increased capacity fade compared to larger DODs (Fig. 4b, for example), but that trend is not very clear after only 12000 hr of cycling since the differences are small (see Fig. S1). In this work, a 25% DOD cell with a 4.06 or 4.20 V UCV discharges down to 3.74 or 3.90 V, respectively—meaning that most of the cell's lifetime is spent at voltages greater than the 100% DOD counterpart which discharges down to 3.0 V and a significant portion of its lifetime will be at lower voltages. It is well known that operating at higher voltages increases the rate of parasitic reactions in the cell,[47,48] so a cell that spends most of its lifetime at relatively higher voltages can be expected to lose more capacity. This can be seen in Figs. 4b–4d where the overall trend shows that smaller DODs result in more capacity fade but is unclear for Fig. 4a at the time this study was concluded. Moreover, Fig. 4 shows that cycling at higher rates results in more capacity fade, which can be rationalized by the fact that C/5 cells will have 10x as many cycles as their C/50 counterpart cycling for the same total time (12000 hr in our case), which can lead to further SEI thickening with every charge.[9]

The capacity loss in this work is showing DOD- and C-rate-dependencies that are in contrast with a similar study from our group by Gauthier et al.,[9] therefore we have summarized the differences between the two studies in Table SV and we explain the origin of these differences in this section. Gauthier's work used polycrystalline NMC622 positive electrode, which demonstrated significant microcracking and positive electrode active mass loss, especially at higher DODs. However, microcracking and positive electrode active

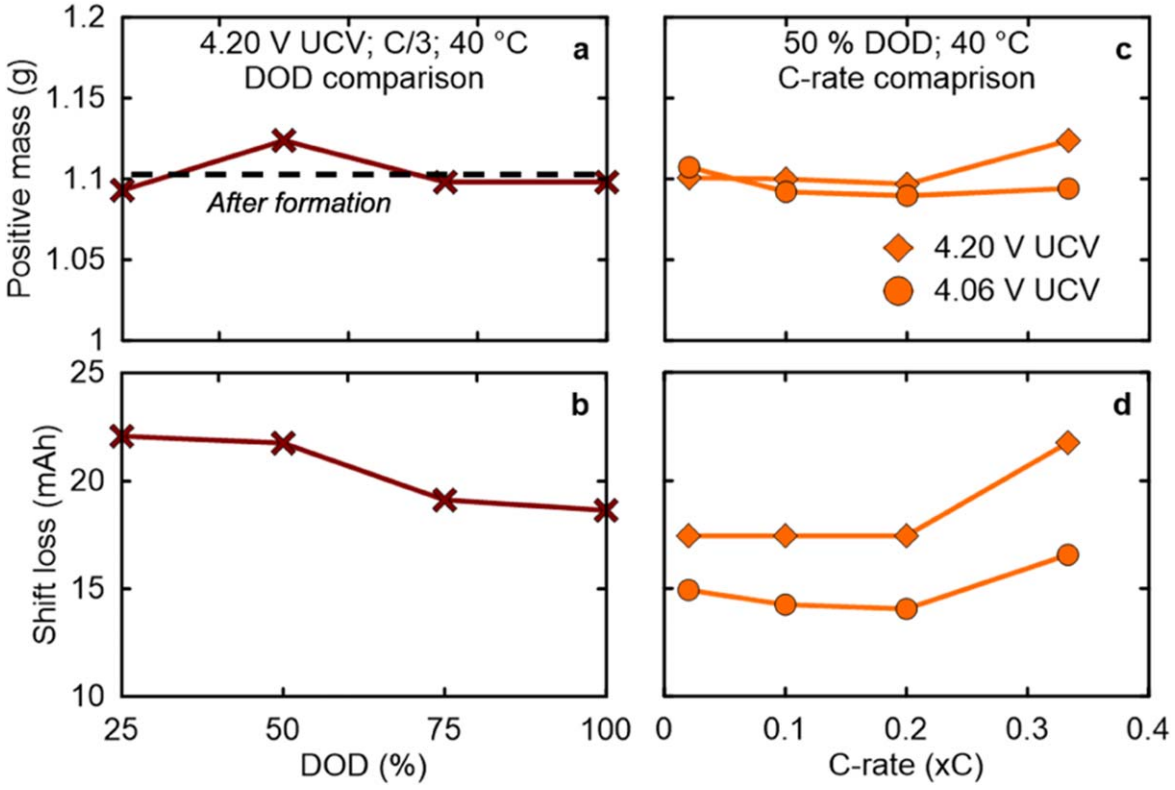


Figure 11. Calculated positive mass (g) and shift loss (mAh) vs (a)–(b) DOD or (c)–(d) C-rate for various SC811/AG cells cycled for 12000 hr. UCV, DOD, c-rate, and/or temperature are indicated in the panels. Dashed line shows the calculated positive electrode active mass after formation (t = 0 hr).

mass loss did not contribute to the observed capacity fade (as shown by differential capacity analysis) since Li loss at the negative electrode was the more dominant mechanism of capacity fade in the cells used by Gauthier et al. Here we used single-crystal NMC811 positive electrode materials which are more robust against microcracking as seen in Figs. 12–13. To reconcile the differences seen in Table SV between this work and Gauthier's, specifically the opposing trends in capacity fade, one must examine the contributions of the different positive electrode and anode materials in both studies to cell failure. NMC622 cycled to 4.10 V in the Gauthier study will not be prone to oxygen release, while NMC811 cells in this work will begin to evolve oxygen when the material is cycled past the remnant of the "H2-H3" plateau as shown by Jung et al.[44] The released oxygen can oxidize the electrolyte[44] and result in the formation of rock-salt or disordered spinel type surface layers.[49] Gauthier's study used natural graphite materials, which are known to perform worse than artificial graphite,[50] and they observed an increase in capacity fade with state of charge range for cells that were cycled with different lower or upper cut-off voltages (see Fig. 16 in Gauthier et al.), which suggests that the graphite side dominates the capacity fade. Exercising the negative electrode over larger SOC ranges, or higher rates, resulted in more fade. In this work we saw the opposite trend: a decrease in capacity fade with increasing DOD, despite the more severe contraction/expansion of graphite particles expected at larger DODs. Additionally, our capacity fade at 12000 hr (Fig. 4) was very similar to that reported at 12000 hr in Fig. 16 of Gauthier et al. despite our use of single-crystal positive electrode and superior artificial graphite. We believe the similar magnitude of capacity fade stems from the fact that we are using NMC811 material which is prone to oxygen release at the upper cut-off voltages used in this work, thus resulting in electrolyte oxidation and Li consumption through some of the well-known pathways reported in the literature.[51,52] Therefore, NMC811/AG cells cycled at lower DODs (which spent more time at relatively higher states of charge) could generate more alkoxides/unwanted electrolyte degradation products, resulting in more capacity fade compared to larger DODs.

Despite some clear differences in capacity retention for 4.06 V cells cycled at different DODs and C-rates (Fig. 4b), Fig. 6 shows that there is no C-rate- or DOD-dependency on the R_{CT} growth after 1 year of cycling. All the 40 °C, 4.06 V, cells in Fig. 6 showed a similar magnitude of R_{CT} increase after cycling compared to the formation data. This is consistent with the cycling data in Figs. 3 and 5 where the 4.06 V cells all showed an equally small increase in ΔV. Interestingly, the 4.20 V cells in Fig. 6 are showing both a C-rate- and DOD-dependence, where R_{CT} increases with DOD and C-rate (see the C/3 column and 100% DOD row in Fig. 6, for example). The increase in R_{CT} with C-rate can be attributed to an increase in cycle number at faster rates, where the cells will operate at higher voltages more frequently leading to an increase in the rate of parasitic reactions, oxygen release and rock-salt surface layer formation, and subsequent impedance growth. However, if R_{CT} was solely dependent on cycle number, then the R_{CT} growth for the 100% DOD cells should be smaller than 25% DOD ones, which is not the case in Fig. 6 for 4.20 V cells, so there must be a more complex interplay between DOD and R_{CT} in this work. Cells cycling at 100% DOD utilize more Li with every charge/discharge which in principle could result in SEI cracking/restructuring[53,54] resulting in an increased rate of parasitic reactions and a higher R_{CT}, compared to the 25% DOD cells. It is possible that cycling at higher DODs resulted in the generation of cross-talk species on the negative electrode that could migrate to the positive electrode and increase R_{CT}, while the lower DOD cells did not generate these species since the graphite volume change was very small at lower DODs and thus the low DOD cells had a smaller R_{CT} increase.

Physical and chemical changes to the electrodes and electrolyte.—We employed various techniques in this work to examine physical and chemical changes to the electrodes and electrolyte of NMC811/AG cells cycled at different conditions. Firstly, we used

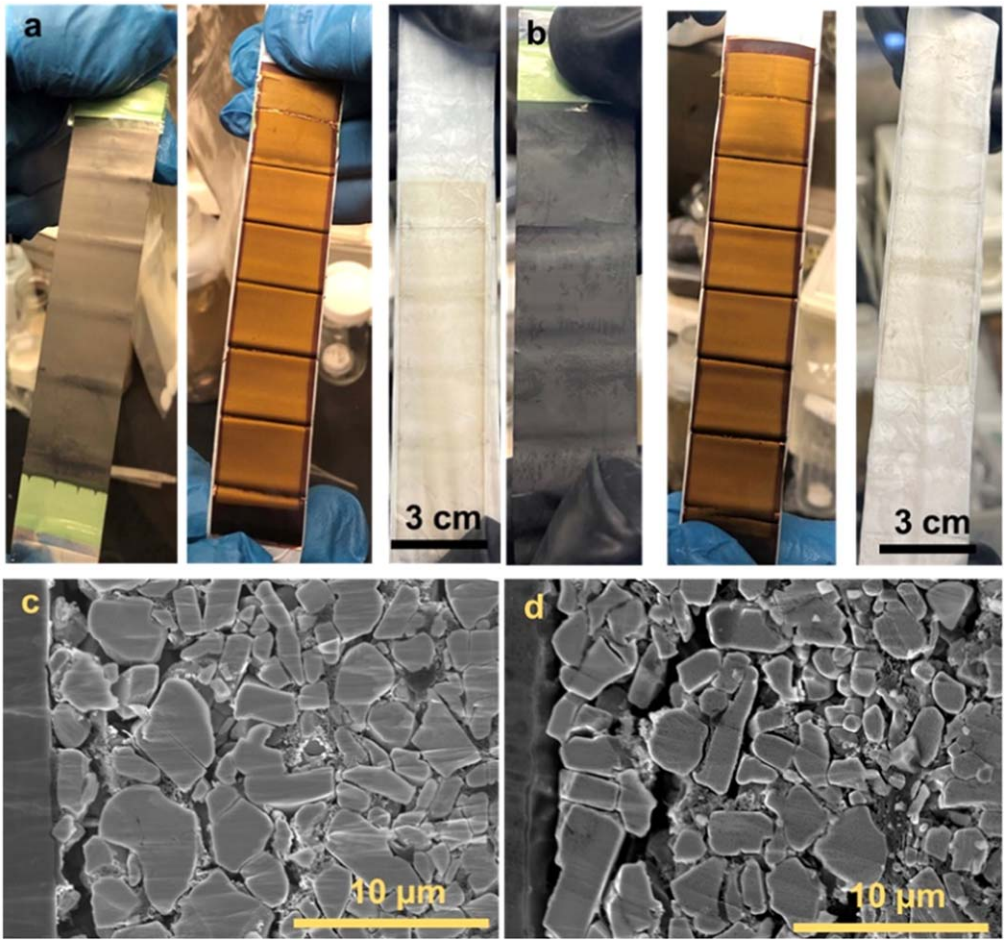


Figure 12. Images of dissembled cells cycled to (a) 4.06 or (b) 4.20 V UCV at C/5 and 25 % DOD at RT, as well as cross-section SEM of the positive electrode taken from the (c) 4.06 or (d) 4.20 V cells. All cells were disassembled at top of charge in an Ar glovebox, and cross-section SEM was done on the delithiated positive electrode.

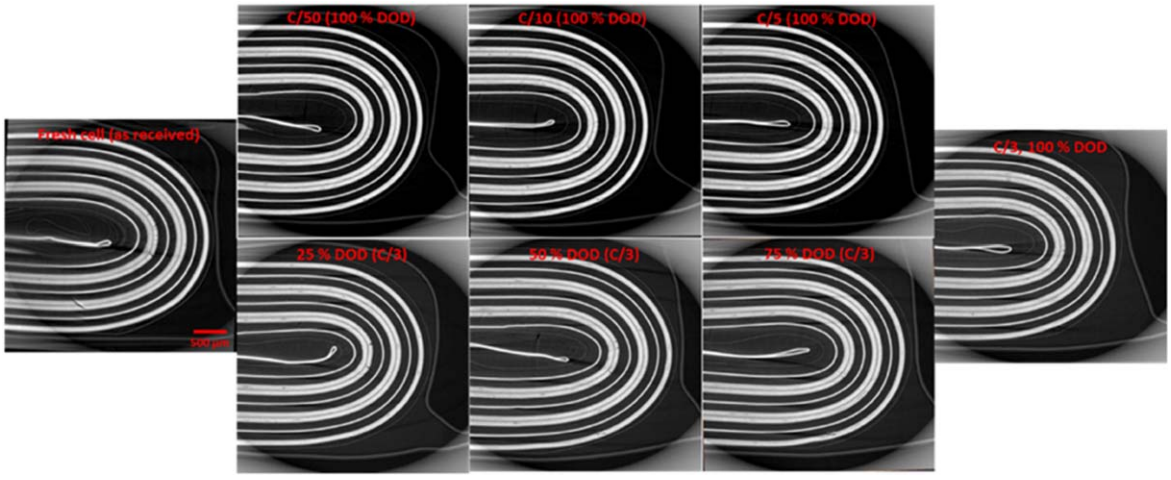


Figure 13. 2D cross sections taken from CT scans of NMC811/AG cells showing a region in the bottom corner of the jelly roll. All cells were cycled at 40 °C to a UCV of 4.20 V for one year. The C-rates and DODs are indicated in each panel. A dry cell with no electrolyte was scanned as received for comparison. The bright and dark bands are the positive and negative electrode, respectively.

XRF to look for TMs dissolved from the positive electrode that migrate and deposit on the negative electrode. Figure 7 shows that the levels of Ni, Mn, and Co on the negative electrodes of 4.06 and 4.20 V cells cycled at 40 °C were very similar to cells that did a formation cycle only. Previous XRF work in our group showed that, in general, TM dissolution and deposition on the negative electrode is not a significant issue in excellent NMC cells.[33,34,55] For example, Thompson et al. examined TM deposition in NMC532 and NMC622 cells that were cycled or stored at 40 or 55 °C and showed that the concentration of TMs on the negative was no more than 0.15% of the total amount of TM in the positive electrode in the worst case. [33] The cells in Thompson et al. were operating at 4.30 or 4.40 V while our

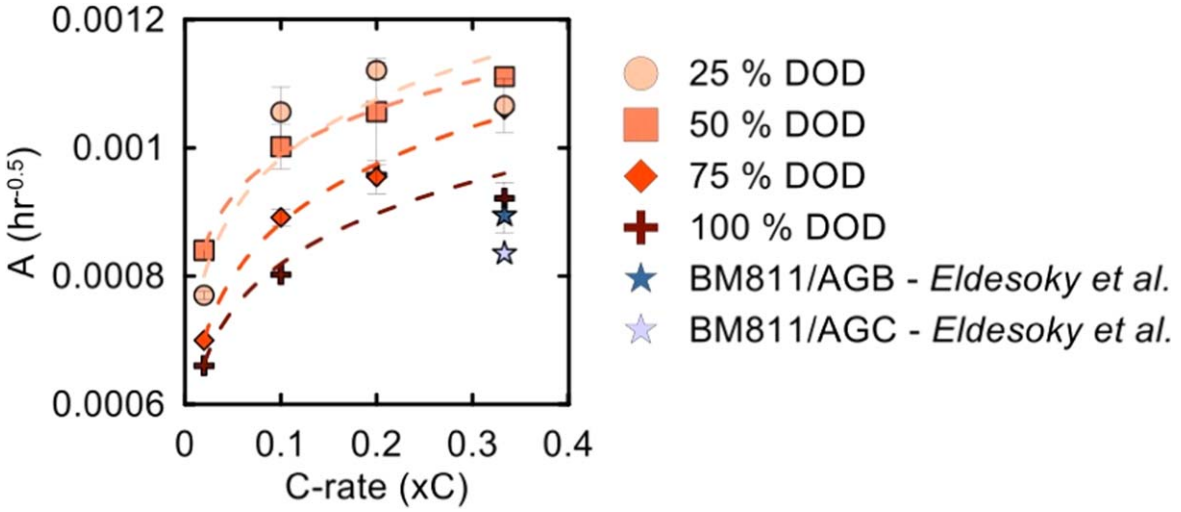


Figure 14. A parameter (hr$^{-0.5}$) vs C-rate for 4.06 V cells cycled at 40 °C at different DODs. Comparative data points from our previous work (C/3, 100% DOD) are shown to highlight the role of graphite material on the rate of SEI growth.[23]

NMC811 cells here did not exceed 4.20 V. While it possible that some small amount of TMs did dissolve in the electrolyte and/or migrated to the anode, we do not believe that this is a significant concern in this work. Additionally, Figs. 11–13 showed no evidence of positive electrode mass loss or particle microcracking, consistent with the absence of TMs on the negative electrode.

Secondly, we used DTA and ultrasonic mapping to study electrolyte changes in the cells. Figure 8 showed no signs of electrolyte unwetting resulting from increased electrode porosity, suggesting that the jelly roll remained sufficiently wetted with electrolyte throughout the course of this study. The cells in this work also exhibited no gassing during cycling (not shown) and no drastic increase in thickness as seen from the CT scans in Figs. 13 or S15 which could have resulted in electrolyte unwetting according to Gauthier et al.[9,27] More detailed discussion of the CT images, and comparison to other NMC types, can be found in a recent work by Bond et al. which used some of the cells in this work.[27]

Finally, differential capacity analysis was used to understand the impact of DOD and C-rate on the cell failure mechanism—i.e., Li inventory loss ("shift loss"), active material loss, or a combination of the two. Figure 11 summarizes the results of the differential capacity analysis, where no positive mass loss was seen—consistent with the XRF, SEM, and CT scans shown above. Furthermore, we saw that cycling at rates of C/5 or slower resulted in similar degrees of shift loss, while the C/3 cells had 2–5 mAh more loss. Similarly, 25–50% DOD cells showed more shift loss compared to 75–100% DOD cells cycled to 4.20 V UCV, but the differences were very small overall (Fig. 11b). The cells in this work performed very well and were similar after 12000 hr of cycling (see Figs. 2–4 and S1), so such subtle differences may be difficult to differentiate with differential capacity analysis. However, we are confident that the calculated shift loss shown in Fig. 11 is consistent with the actual capacity loss of the cells as shown in Fig. S14. The readers will note that the capacity loss values shown in Fig. S14 are smaller than that shown for some of the cells in Fig. 4. The cells in this work have a 1 mm anode overhang for safety reasons (which is common in commercial cells) where some Li can diffuse and intercalate in the graphite during cycling. All cycles used in differential capacity analysis were obtained on Dalhousie's ultra-high precision charger. The cells rested for about 3 weeks, during which additional post-cycling tests were carried out, followed by the test protocol described the experimental section. The 3-week wait plus the 12 hr hold at bottom of discharge will result in the removal of Li from the anode overhang to be re-intercalated into the positive electrode. Then this recovered lithium can be utilized in the subsequent cycles, resulting in the increased capacity seen in Fig. S14 compared to Fig. 4.

SEI Growth and cells with improved graphite.—We showed that the most impactful change to the cycling conditions of NMC811/AG cells is to limit the UCV to 4.06 V, which virtually eliminates ΔV growth and significantly improves capacity retention (and the total energy output of the cell due to the longer lifetime at 4.06 vs 4.20 V) as shown in Figs. 2–5. Further, modeling the capacity fade of 4.06 V UCV cells (where capacity loss is only due to Li inventory loss in the negative electrode SEI) using square-root time model of SEI growth shows that fast rates and *smaller* DODs result in more rapid SEI growth as shown in Fig. 14. The lower the DOD, the more time the cell spends at relatively higher voltages, which could result in the generation of cross-talk species or oxygen release in some cases, while gaining more cycles over a fixed timeframe. Therefore, a balance between C-rate and DOD must be considered depending on the intended applications, and careful thought should be given to the role that the positive and negative electrodes play in cell failure as discussed above.

Comparing the results in Fig. 14 to our previous work on NMC811 cells with improved graphite materials, one can clearly see the benefit of using the so-called "AGC" artificial graphite material which exhibits slower SEI growth rate and a lower electrochemically accessible surface for parasitic reactions.[23] Figure 15 highlights the impact of the UCV and using "AGC" graphite at 70 and 40 °C, where the choice of 70 °C dramatically accelerates cell failure and enables faster screening of cell performance under different cycling conditions.[24] The remarkable ΔV growth shown in Fig. 15e as a function of UCV should convince the reader of the importance of limiting the cell's UCV and using the best graphite material available to yield the longest lifetime possible (see Fig. 15c). While limiting the UCV is counterintuitive given the current push in academia and industry for higher energy densities, the scarcity of battery raw materials and urgent demand for more battery manufacturing can be addressed by favouring lifetime over energy density until we can develop commercially-viable strategies to push the cell upper cut-off voltage without compromising lifetime. Furthermore, since it takes 50 to 60 kWh to produce 1 kWh of Li-ion cells[55] it is paramount to extend the lifetime and cycle life of Li-ion cells as much as possible to avoid excess use of energy in battery manufacturing.

Conclusions

This work examined the impact of DOD, C-rate, UCV, and temperature on the lifetime of NMC811/AG pouch cells. After cycling, cells underwent various chemical and physical characterization tests to study the impact of cycling conditions on electrode

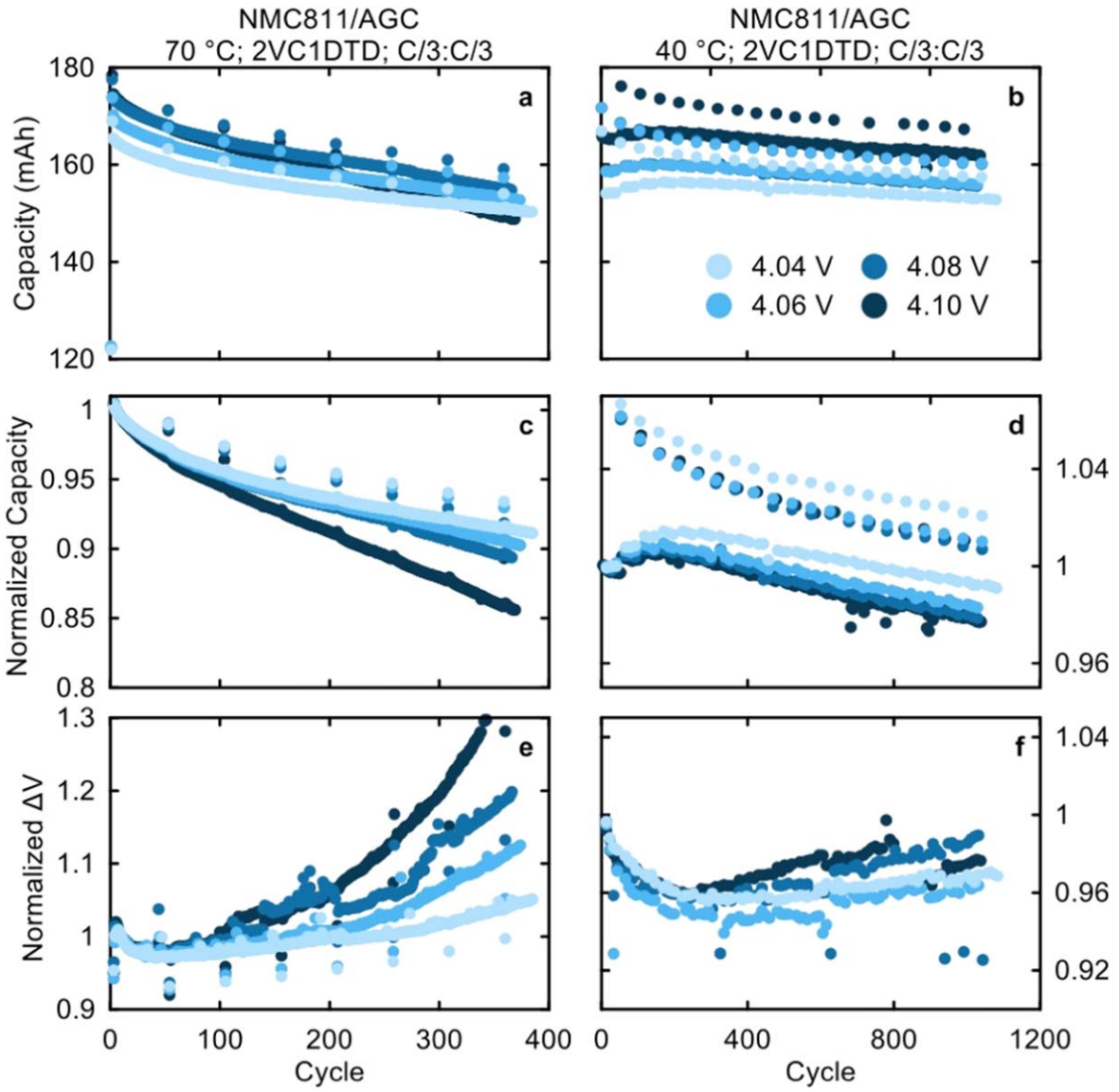


Figure 15. (a)–(b) capacity (mAh), (c)–(d) normalized capacity, and (e)–(f) normalized ΔV for NMC811/AGC cells cycled to different UCVs at 70 and 40 °C with the same electrolyte as NMC811/AG cells in this work. Similar to the NMC811/AG results shown throughout this work, this figure highlights the importance of limiting the UCV even when more competitive graphite materials are used.

and electrolyte changes. The main findings of this work can be summarized as follows:

1. Capacity loss and ΔV growth generally increased with C-rate;
2. Overall, the impact of DOD on capacity loss and ΔV growth was small, but we saw a slight increase in capacity loss with smaller DODs which we attributed to the fact that the cells spent more time at higher voltages when the DODs were small;
3. EIS measurements at 50% SOC and 10 °C showed that R_{CT} increased with DOD and decreased with C-rate for 4.20 V cells cycled at 40 °C, but no C-rate- or DOD-dependency for their 4.06 V counterparts;
4. XRF mapping of the negative electrode after 1 year of cycling at 40 °C showed no significant TM concentrations for 4.06 or 4.20 V cells;
5. Ultrasonic mapping and DTA showed no significant electrolyte unwetting or C-rate-/DOD-dependant chemical changes, respectively, for 4.20 V cells cycled at 40 °C;
6. Differential capacity analysis and cross-section SEM of the positive electrode showed no signs of positive electrode mass loss or particle microcracking, consistent with the robustness of the single crystal NMC811 used here;
7. Synchrotron CT showed no pronounced thickness increases for 4.20 V cells cycled at 40 °C, and no signs of electrode damage or delamination;
8. Rate of SEI growth increases with higher rates and smaller DODs for 4.20 V UCV cells cycled at 40 °C;
9. Using improved graphite materials and lowering the UCV below 4.06 V can further enhance cell lifetime;
10. Maximizing cell lifetime, even at the expense of initial cell energy, is crucial to meeting the demand for battery capacity while facing a global shortage of the scarce materials need to produce Li-ion cells.

We believe the cells used in this work are near the state-of-the-art for NMC811/AG cells and we hope this study will provide useful guidelines for determining the cycling conditions needed to prolong the lifetime of NMC811 further. Future work will focus on the impact of DOD and C-rate on cells with different grades of single crystal NMC, as well as LFP and NMC811 cells with different graphite materials.

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ORCID

A. Eldesoky https://orcid.org/0000-0002-1501-5139
T. Bond https://orcid.org/0000-0003-2043-0094
Nicholas Kowalski https://orcid.org/0000-0001-7907-4448
D. Rathore https://orcid.org/0000-0003-2909-7726
J. R. Dahn https://orcid.org/0000-0002-6997-2436

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